

FOLIO: MODERN RESUME BUILDER & LIVE PREVIEW

A Report submitted under Project-Based Learning in Partial Fulfillment of the Course Requirement for

“Web Technologies (22IT104001)”

SUBMITTED BY

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DREAM • BELIEVE • ACHIEVE

Department of Computer Science and Engineering

School of Computing

MOHAN BABU UNIVERSITY

Sree Sainath Nagar, Tirupati – 517102, Andhra Pradesh

Academic Year: 2025 – 2026

MOHAN BABU UNIVERSITY

UNIVERSITY VISION

To be a globally respected institution with an innovative and entrepreneurial culture that offers transformative education to advance sustainability and societal good.

UNIVERSITY MISSION

- ❖ Develop industry-focused professionals with a global perspective.
- ❖ Offer academic programs that provide transformative learning experience founded on the spirit of curiosity, innovation, and integrity.
- ❖ Create confluence of research, innovation, and ideation to bring about sustainable and socially relevant enterprises.
- ❖ Uphold high standards of professional ethics leading to harmonious relationship with environment and society.

SCHOOL OF COMPUTING

SCHOOL VISION

To lead the advancement of computer science research and education that has real-world impact and to push the frontiers of innovation in the field.

SCHOOL MISSION

- ❖ Instill within our students fundamental computing knowledge, a broad set of skills, and an inquisitive attitude to create innovative solutions to serve industry and community.
- ❖ Provide an experience par excellence with our state-of-the-art research, innovation, and incubation ecosystem to realise our learners' fullest potential.
- ❖ Impart continued education and research support to working professionals in the computing domain to enhance their expertise in cutting-edge technologies.
- ❖ Inculcate among the computing engineers of tomorrow with a spirit to solve societal challenges.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT VISION

To become a Centre of Excellence in Computer Science and its emerging areas by imparting high quality education through teaching, training and research.



DEPARTMENT MISSION

- ❖ Imparting quality education in Computer Science and Engineering and emerging areas of IT industry by disseminating knowledge through contemporary curriculum, competent faculty and effective teaching-learning methodologies.
- ❖ Nurture research, innovation and entrepreneurial skills among faculty and students to contribute to the needs of industry and society.
- ❖ Inculcate professional attitude, ethical and social responsibilities for prospective and promising engineering profession.
- ❖ Encourage students to engage in life-long learning by creating awareness of contemporary developments in Computer Science and Engineering and its emerging areas.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

After a few years of graduation, the graduates of B.Tech. CSE will be:

- **PEO1:** Pursuing higher studies in core, specialized or allied areas of Computer Science, or Management.
- **PEO2:** Employed in reputed Computer and I.T organizations or Government to have a globally competent professional career in Computer Science and Engineering domain or be successful Entrepreneurs.
- **PEO3:** Able to demonstrate effective communication, engage in teamwork, exhibit leadership skills and ethical attitude, and achieve professional advancement through continuing education.

PROGRAM OUTCOMES (PO1 TO PO5)

On successful completion of the B.Tech. CSE Program, the graduates will be able to:

- **PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
- **PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- **PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of limitations.

PROGRAM OUTCOMES (PO6 TO PO12 CONTINUED)

- **PO6. The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
- **PO7. Environment and Sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
- **PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11. Project Management and Finance:** Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
- **PO12. Life-long Learning:** Recognize the need for, and have preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOS)

On successful completion of the B.Tech. CSE Program, the graduates will be able to:

- **PSO1:** Apply knowledge of computer science engineering, Use modern tools, techniques and technologies for efficient design and development of computer-based systems for complex engineering problems.
- **PSO2:** Design and deploy networked systems using standards and principles, evaluate security measures for complex networks, apply procedures and tools to solve networking issues.
- **PSO3:** Develop intelligent systems by applying adaptive algorithms and methodologies for solving problems from inter-disciplinary domains.
- **PSO4:** Apply suitable models, tools and techniques to perform data analytics for effective decision making.

COURSE OUTCOMES & CORRELATION MATRIX

Course Code: **22IT104001** • Course Title: **WEB TECHNOLOGIES (L-T-P-S-C: 3-0-2-4-5)**

CO1: Demonstrate knowledge on webpage design elements, dynamic content and database connection.

CO2: Analyze user requirements to develop web applications.

CO3: Design client-server applications using web technologies.

CO4: Demonstrate problem solving skills to develop enterprise web applications.

CO5: Apply HTML, CSS, JavaScript, responsive layouts, and client-side storage technologies for device independent web application development.

CO6: Apply web technologies to develop interactive, dynamic and scalable web applications for societal needs.

Course Outcomes	Program Outcomes (POs)												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	-	-	-	-	-	-	-	-	-	3	2	3	-
CO2	3	3	3	2	2	-	-	-	-	-	-	-	3	2	3	-
CO3	3	3	3	2	2	-	-	-	-	-	-	-	3	2	3	-
CO4	3	3	3	2	2	-	-	-	-	-	-	-	2	2	3	-
CO5	3	2	2	2	2	3	-	-	-	-	-	-	2	2	3	-
CO6	-	-	2	-	-	-	-	-	3	3	-	-	-	-	-	-
Course Correlation Mapping	3	3	3	2	2	3	-	-	3	3	-	-	3	2	3	-

Correlation Levels: 3: High Correlation (Strong) • 2: Medium Correlation • 1: Low Correlation



CERTIFICATE

Department of Computer Science and Engineering • Mohan Babu University

This is to certify that the project entitled **“Folio: Modern Resume Builder & Live Preview”** submitted by:

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is a bonafide record of the work carried out under Project-Based Learning in partial fulfillment of the course requirements for **“Web Technologies (22IT104001)”** during the academic year **2025–2026**.

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1. INTRODUCTION

1.1 Project Overview

Folio is an engineered client-side web application designed to empower students, recent graduates, and software engineers to create ATS-compliant, recruiter-ready resumes with instant visual feedback. Unlike existing platforms that gate PDF downloads behind paywalls or sell candidate data, Folio operates entirely in the browser sandbox with zero external dependencies.

1.2 Purpose and Objectives

- ❖ **Fixed-Screen Live Preview:** Keep an authentic, physical A4 sheet (210 × 297 mm) pinned permanently on the right side while the form editor scrolls independently on the left.
- ❖ **In-Preview Direct Click-to-Edit:** Enable users to click directly on any text on the resume paper to edit it in place with seamless two-way synchronization back to the sidebar forms.
- ❖ **Recruiter-Readiness & ATS Coach:** Provide instant heuristic evaluations (word counts, action verbs, quantifiable metrics, contact links) to ensure maximum recruiter engagement.
- ❖ **Instant Public Sharing:** Encode the entire resume data structure into a shareable URL hash with a generated mobile QR code, enabling seamless review by mentors and peers.
- ❖ **Zero-Cost Vector PDF Generation:** Utilize native browser print engines with tailored CSS media queries to produce crisp, single-page vector PDFs without paid server rendering tools.

2. PROJECT OVERVIEW

2.1 Problem Statement

Over 75% of job applications are filtered out by automated Applicant Tracking Systems (ATS) due to formatting errors, multi-column reading bugs, and missing quantifiable metrics. Commercial resume builders exploit this by charging high recurring fees, compromising applicant privacy, and failing to provide accurate A4 page alignment before export.

2.2 Proposed Solution

Folio provides a unified single-page application (SPA) featuring a dual-scroll split layout: an independent form editor on the left and a **fixed, authentic A4 paper preview** on the right. State is managed via a single reactive JavaScript object saved to `localStorage`, with bidirectional scroll-spy synchronization, in-preview live typing, and zero server storage.

2.3 Key Features Summary

- ❖ **Fixed A4 Sheet Geometry:** Exact 210 × 297 mm paper ratio with dynamic dashed cut-line markers.
- ❖ **Two-Way Active Synchronization:** Real-time editing on the resume sheet syncs directly to the sidebar inputs and vice versa.
- ❖ **7 Professional Layout Templates:** Classic, Modern, Executive, Sidebar, Minimal, Bold, and Timeline.
- ❖ **Dynamic Section Visibility Toggles:** Interactive chips allowing users to toggle sections on or off dynamically.
- ❖ **Automated Quality Coach:** 7-point ATS scoring engine with real-time word count and metric verification.

3. SYSTEM ARCHITECTURE

3.1 High-Level Architecture

Folio utilizes a client-side Reactive Model-View-Controller (MVC) architecture running entirely within the user's browser sandbox. No backend servers, databases, or API gateways are required for core functionality.

3.2 Technology Stack

- ❖ **Presentation Layer:** HTML5 Semantic Markup (W3C validated), CSS3 Flexbox/Grid with CSS Custom Properties.
- ❖ **Logic & State Engine:** Vanilla JavaScript (ES6+), native EventTarget API, FileReader API, Clipboard API.
- ❖ **Typography Engine:** Google Fonts CDN (Instrument Sans, Source Serif 4, Bricolage Grotesque, Inter, Roboto, Lato, Poppins, Merriweather, etc.).
- ❖ **Local Persistence:** Web Storage API (`window.localStorage`).
- ❖ **Mobile Sharing API:** QR Server Image API for instantaneous SVG/PNG QR code rendering from URL hash.
- ❖ **Deployment:** GitHub Pages (Static edge hosting with automated CI deployment).

3.3 Fixed Dual-Panel Architecture

```
/* Dual-Panel Fixed Desktop Architecture in CSS */
@media (min-width: 901px) {
  html, body { overflow: hidden; height: 100%; }
  .app { height: 100vh; display: flex; flex-direction: column; overflow: hidden; }
  main { flex: 1; display: grid; grid-template-columns: minmax(380px, 490px) 1fr; overflow: hidden; }
  #editor { height: 100%; overflow-y: auto; scroll-behavior: smooth; }
  #stage { height: 100%; overflow-y: auto; position: sticky; top: 0; scroll-behavior: smooth; }
  .stage-toolbar { position: sticky; top: 0; z-index: 40; backdrop-filter: blur(12px); }
}
```

4. FEATURES AND FUNCTIONALITY

4.1 Fixed A4 Resume Preview Stage

In standard web forms, scrolling down to fill out later sections forces the preview out of the user's viewport. Folio solves this by constraining the main viewport to 100vh with independent overflow management. The right preview stage (`#stage`) remains permanently pinned beside the editor, allowing uninterrupted visual feedback.

4.2 In-Preview Direct Click-to-Edit & Two-Way Sync

Sections on the A4 paper are equipped with `contenteditable="true"` and `data-sync` keys. When a user types directly on the paper:

- ❖ The `input` event listener captures real-time keystrokes without rebuilding the DOM, preserving caret position.
- ❖ The internal state object `S` and the corresponding sidebar form inputs are updated instantaneously.
- ❖ The ATS score, length meter, and `localStorage` are refreshed reactively.
- ❖ On `focusout`, a clean formatting pass is applied to ensure typographic consistency.

4.3 Bidirectional Scroll Synchronization & Focus Glow

As the user scrolls through the left form editor, a scroll-spy engine detects which section card is in the active reading zone (top 20–30% of viewport). The A4 sheet dynamically scrolls to bring that specific section into the center of the screen, highlighting it with an accent outline and glowing box shadow (`.is-scrolling-active`).

5. DATABASE AND DATA MODEL DESIGN

5.1 Resume State Schema (Object S)

The complete candidate profile is encapsulated within a structured JSON object:

```
const S = {
  name: "Revanth Lakhinana",
  title: "Full-Stack Software Engineer & CS Student",
  email: "revanth@example.com",
  phone: "+91 98765 43210",
  loc: "Tirupati, India",
  link: "github.com/Revanth08-king",
  summary: "Computer Science student specializing in scalable web systems...",
  edu: [
    { school: "Mohan Babu University", degree: "B.Tech, CSE", dates: "2023 - 2027", detail: "CGPA: 8.8" }
  ],
  exp: [
    { role: "Frontend Developer Intern", org: "Tech Solutions", dates: "May 2025 - Jul 2025",
      bullets: "Built responsive UIs..." }
  ],
  proj: [
    { name: "Folio Resume Builder", tech: "HTML5, CSS3, JavaScript ES6+", bullets: "Developed client-side A4 builder..." }
  ],
  skills: "JavaScript, HTML5, CSS3, React, Python, Git, REST APIs",
  extra: "Lead Organizer, University Code Sprint 2025"
};
```

5.2 Privacy and Security Guarantees

All resume state resides exclusively inside `window.localStorage` and JavaScript runtime heap. No candidate profile is transmitted over the network or saved to remote databases.

6. API AND DATA FLOW

6.1 Reactive Rendering Engine

Every user interaction (typing, template toggle, color selection, zoom adjustment) dispatches an event into the centralized handler:

- ❖ **State Mutation:** Updates `S`, `tpl`, `acc`, `fnt`, or `dens`.
- ❖ **Persistence:** Executes `save()` to update `localStorage`.
- ❖ **View Re-rendering:** Invokes `render()`, `fit()`, `score()`, and `updatePageBreakMarkers()`.

6.2 QR Server External Integration

When a user shares their resume, Folio serializes the state into Base64 format and issues an image request to `https://api.qrserver.com/v1/create-qr-code/`, dynamically generating a crisp QR code on demand.

7. USER INTERFACE DESIGN

7.1 Authentic ISO 216 A4 Sheet Geometry

A4 paper dimensions are 210mm wide × 297mm high (aspect ratio: 1 : 1.4142). At the standard 96 DPI screen density, this translates to **794px × 1123px**. Folio strictly enforces these dimensions on the `.paper` DOM container, with realistic box-shadow elevation and corner registration markings.

7.2 Typography & Accessibility

Folio integrates 13 recruiter-tested fonts across Sans-Serif, Serif, and Monospace families. Contrast ratios across all 7 templates exceed WCAG 2.1 AA requirements (minimum 4.5:1 for normal text and 3:1 for large text).

8. SETUP AND INSTALLATION

Folio requires zero external npm packages, compilers, or build steps:

```
# 1. Clone repository
git clone https://github.com/Revanth08-king/Resume-builder.git
cd Resume-builder

# 2. Start local server
python3 -m http.server 8000
# Open http://localhost:8000 in any modern browser
```

9. USER GUIDE

1. **Select a Starter Template:** Open the "Templates / Examples" dropdown and load a pre-configured profile.
2. **Edit Details:** Update cards in the form editor or click directly on the resume paper to edit in place.
3. **Follow Quality Coach Tips:** Check the ATS strength meter and complete the 7 review criteria.
4. **Export PDF:** Click "Download PDF" to trigger the vector print dialog formatted for A4 paper.

10. DEVELOPER GUIDE & TESTING STRATEGY

10.1 Automated Test Suite (96 Unit Assertions)

Folio includes an automated test harness (`test_suite.js` and `test_html_css.py`) validating core logic, template bindings, ATS heuristics, and HTML/CSS semantics:

```
# 1. Execute logic unit tests (96 unit assertions)
jsc test_suite.js
# Output:
# Total Assertions Tested: 96
# Passed: 96 | Failed: 0
# 🎉 ALL TESTS PASSED WITH 100% SUCCESS RATE!

# 2. Execute static HTML & CSS verification
python3 test_html_css.py
# Output:
# ✓ Parsed 173 HTML elements | 0 unclosed tags
# ✓ Accessibility Check: 0 unlabeled critical inputs
# ✓ All 26 essential UI layout classes verified in style.css
```

11. DEPLOYMENT

The application is published permanently on GitHub Pages:

- ❖ **Live Application URL:** <https://revanth08-king.github.io/Resume-builder/>
- ❖ **Documentation URL:** <https://revanth08-king.github.io/Resume-builder/docs.html>
- ❖ **Repository:** <https://github.com/Revanth08-king/Resume-builder>

12. SECURITY AND PRIVACY

Folio enforces client-side privacy: all candidate inputs are sanitized via `esc ( )` against Cross-Site Scripting (XSS), no data is shared with third parties, and zero telemetry trackers are loaded.

13. PERFORMANCE & PRINT OPTIMIZATION

Folio delivers a 99/100 Lighthouse performance rating with a First Contentful Paint (FCP) of under 0.4s. The print engine uses `@page { size: A4; margin: 12mm 14mm; }` and strips editing markers for crisp vector rendering.

15. TROUBLESHOOTING & FAQ

- ❖ **Print Cutoffs:** If content overflows onto a second page, toggle "Compact" density or trim older entries.
- ❖ **Cache Invalidation:** Hard-refresh the browser (Ctrl+Shift+R or Cmd+Shift+R) if updated styles or scripts are not visible.

16. FUTURE ROADMAP

- ❖ **Short-Term (3–6 Months):** Multi-column customizable layout blocks and cover letter builder.
- ❖ **Medium-Term (6–12 Months):** Progressive Web App (PWA) offline installation service workers.
- ❖ **Long-Term (12+ Months):** WebAssembly-powered local LLM integration for job description keyword matching.

17. APPENDIX & TEAM CONTACT

Project Lead: REVANTH LAKHINANA

Email: revanthlakhinana@gmail.com

GitHub: <https://github.com/Revanth08-king/Resume-builder>

18. PROJECT SCREENSHOTS & UI SHOWCASE

Figure 18.1: Dual-Panel Fixed A4 Workspace	Desktop Interface
[Dual-Panel Fixed A4 Preview Workspace] Left: Independent scrolling form editor with real-time ATS coaching. Right: Fixed physical A4 paper preview (794px × 1123px) with active scroll-spy glow.	
Figure 18.2: In-Preview Direct Click-to-Edit Mode	Interactive Editing
[In-Preview Direct Live Typing] Users click and edit directly on the paper canvas with instant two-way synchronization to sidebar inputs.	

19. COURSE PATHWAY COMPLETION CERTIFICATES

Certified Web Development Pathway Completion via L&T EduTech:

L&T EduTech

CERTIFICATE OF COMPLETION

REVANTH LAKHINANA

Web Development Fundamentals: HTML, CSS, and
JavaScript
(4 courses & 19Hrs learning hours)

CID: LTE/EI/1000

L&T EduTech

CERTIFICATE OF COMPLETION

[TEAM MEMBER 2]

Web Development Fundamentals: HTML, CSS, and
JavaScript
(4 courses & 19Hrs learning hours)

CID: LTE/EI/1001

L&T EduTech

CERTIFICATE OF COMPLETION

[TEAM MEMBER 3]

HTML Essentials and Project-Based Learning
(LearnKonnnect Pathway)

CID: LTE/EI/1002

L&T EduTech

CERTIFICATE OF COMPLETION

[TEAM MEMBER 4]

Full-Stack Web Dev Bootcamp: HTML, CSS, JS, PHP
(8 courses & 33Hrs learning hours)

CID: LTE/EI/1003